Hui Hui Xie

Dingxin Postdoctoral Researcher | College of Physics, Jilin University

huihuixie@jlu.edu.cn | Changchun, China

xhh22900.github.io | ORCID: 0000-0002-6185-0856

Research overview

14 published journal articles, including seven as first author. First-author publications include a *Physical Review C* Letter. One first-author manuscript is under review at *Physical Review Letters*.

Research interests

Nuclear charge distributions, radii, and higher radial moments within relativistic (covariant) density functional theory.

Nuclear-structure and finite-nuclear-size effects in precision atomic physics: hydrogenlike ions, muonic atoms, bound-electron g factors, and electron scattering.

Earlier work in precision atomic calculations and quantum walks; interest in connections to quantum information and numerical quantum dynamics.

Appointments and education

Dec 2024 - present	**Dingxin Postdoctoral Researcher**, Jilin University Advisor: Jian Li
Sep 2021 - Sep 2024	**Ph.D. in Particle and Nuclear Physics**, Jilin University Advisor: Jian Li Thesis: Microscopic studies and applications of nuclear charge properties based on relativistic density functional theory
Sep 2018 - Jun 2021	**Master's degree in Theoretical Physics**, Jilin University Advisors: Li Guang Jiao and Guo-Mo Zeng Quantum information with Guo-Mo Zeng (about two years); precision atomic calculations with Li Guang Jiao (about one year).
Sep 2014 - Jun 2018	**Bachelor's degree in Electronic Science and Technology** Changchun University of Science and Technology

Research funding

China Postdoctoral Science Foundation, General Program (78th batch)
Principal investigator | Grant 2025M783390 | December 2025
Microscopic study of finite-nuclear-size effects in hydrogenlike ions.

Selected research contributions

Nuclear charge radii. Studied intrinsic nucleon electromagnetic structure and quadrupole shape fluctuations to describe calcium-isotope radius evolution without local parameter readjustment [1, 2].

Muonic-atom spectra. Used microscopic charge densities to benchmark model dependence and methods for extracting charge radii, higher radial moments, and low-momentum form factors [3, 4].

Finite nuclear size and g factors. Calculated finite-nuclear-size corrections and assessed nuclear contributions to Ne isotope g-factor differences using deformation and Bayesian inference [5, S1].

Quantum information and precision atomic calculations. Master's work covered quantum walks on Sierpinski fractal networks [7], relativistic corrections for screened hydrogenlike atoms [6], and critical screening behavior [14].

Published journal articles

14 articles, seven as first author. The applicant's name is shown in bold.

First-author articles

[1] **H. H. Xie**, J. Li, Y. L. Yang, and P. W. Zhao, "Charge radii of calcium isotopes within relativistic density functional theory: The finite size of the nucleon and quadrupole shape-fluctuation effects," Phys. Rev. C **112**, L021303 (2025) (Letter).

[2] **H. H. Xie** and J. Li, "Impact of intrinsic electromagnetic structure on the nuclear charge radius in relativistic density functional theory," Phys. Rev. C **110**, 064319 (2024).

[3] **H. H. Xie**, J. Li, and H. Liang, "Extraction of higher-order radial moments of nuclear charge density from muonic atom spectroscopy," Phys. Rev. C **109**, 034309 (2024).

[4] **H. H. Xie**, T. Naito, J. Li, and H. Liang, "Revisiting the extraction of charge radii of ^{40}Ca and ^{208}Pb with muonic atom spectroscopy," Phys. Lett. B **846**, 138232 (2023).

[5] **H. H. Xie**, J. Li, L. G. Jiao, and Y. K. Ho, "Finite-nuclear-size effect in hydrogenlike ions with relativistic nuclear structure," Phys. Rev. A **107**, 042807 (2023).

[6] **H. H. Xie**, L. G. Jiao, A. Liu, and Y. K. Ho, "High-precision calculation of relativistic corrections for hydrogen-like atoms with screened Coulomb potentials," Int. J. Quantum Chem. **121**, e26653 (2021).

[7] **H. H. Xie** and G. M. Zeng, "Quantum walks on Sierpinski gasket and Sierpinski tetrahedron," Quantum Inf. Process. **20**, 240 (2021).

Coauthored articles

[8] Y. D. Wang, **H. H. Xie**, T. S. Shang, P. X. Du, J. Li, H. Liang, and K. Zhang, "Microscopic study of charge properties in halo nuclei," Phys. Rev. C **114**, 024315 (2026).

[9] D. S. Liu, **H. H. Xie**, P. X. Du, T. S. Shang, J. Li, J. G. Li, and T. Naito, "Finite-nuclear-size effect for hydrogenlike ions under high external pressure," Commun. Theor. Phys. **78**, 075301 (2026).

[10] Y. D. Wang, T. S. Shang, **H. H. Xie**, P. X. Du, J. Li, and H. Liang, "Predictions of charge density distributions for nuclei with $Z \geq 8$," Nucl. Sci. Tech. **37**, 93 (2026).

[11] T. Y. Wu, B. H. Sun, **H. H. Xie**, J. Y. Xu, and G. Guo, "Point-proton density distributions of stable nuclei," At. Data Nucl. Data Tables **165**, 101733 (2025).

[12] T. S. Shang, **H. H. Xie**, J. Li, and H. Liang, "Global prediction of nuclear charge density distributions using a deep neural network," Phys. Rev. C **110**, 014308 (2024).

[13] P. Guo *et al.* (including **H. H. Xie**), "Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even-*Z* nuclei," At. Data Nucl. Data Tables **158**, 101661 (2024).

[14] L. G. Jiao, **H. H. Xie**, A. Liu, H. E. Montgomery Jr., and Y. K. Ho, "Critical screening parameters and critical behaviors of one-electron systems with screened Coulomb potentials," J. Phys. B **54**, 175002 (2021).

Manuscripts and preprints

[S1] **H. H. Xie**, J. Li, B. S. Tu, H. Liang, and X. W. Ma, "Microscopic nuclear structure calculations unlock high-order recoil tests in bound-electron g factors." Submitted to *Physical Review Letters* (under review).

[S2] D. S. Liu, **H. H. Xie**, P. X. Du, J. Li, and T. Naito, "The pseudospectral method for the Dirac equation with confining potential." arXiv:2605.25390 (2026).

Selected oral presentations

Jun 2026	Microscopic Nuclear Structure Calculations Unlock High-Order Recoil Tests in Bound-Electron g Factors Workshop on Frontiers in Nuclear and Atomic Physics, Jilin University
Mar 2024	Extracting Higher-Order Moments of Nuclear Charge Density from Muonic-Atom Spectroscopy 19th National Conference on Nuclear Structure, Ruichang, Jiangxi
Apr 2023	Finite-Nuclear-Size Effects within Covariant Density Functional Theory 12th School on Covariant Density Functional Theory in Nuclear Physics, Tianjin
Aug 2022	Finite-Nuclear-Size Effects within Covariant Density Functional Theory 18th National Conference on Nuclear Structure, Guiyang

Collaboration and graduate research support

Contributed to collaborative studies of nuclear charge-density predictions, point-proton density distributions, the DRHBc nuclear mass table, and halo-nucleus charge properties [8-13]. Assisted the advisor in supervising graduate research on halo-nucleus charge properties and finite-nuclear-size effects in confined hydrogenlike ions, with outcomes including papers [8, 9] and preprint [S2].

Numerical methods and technical skills

Programming and computing. Proficient in Fortran; working proficiency in Python (NumPy, SciPy, PyTorch), C/C++, and MATLAB. Experience with batch and multiprocessing workflows on Linux.

Nuclear-structure calculations. Use RCHB for research and modify or extend its data outputs as needed.

Dirac solvers. Maintain and use high-precision generalized pseudospectral solvers; familiar with the algorithm and kinetic-balance improvements.

Electron scattering. Independently wrote a Dirac partial-wave code for elastic electron-nucleus scattering, with reference to the open-source ELSEPA program.

Quantum dynamics and atomic structure. Simulations of quantum walks on fractal networks, including probability distributions, return probabilities, and standard deviations; precision calculations of screened one-electron systems.

Theory and statistics. Relativistic (covariant) density functional theory, Bayesian inference (MCMC), and machine-learning uncertainty quantification.